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Method and apparatus for deposition of piezo-electric materials

Abstract

Examples disclosed herein relate to an apparatus and method of forming thin film layers on a substrate. A first piezoelectric material layer is deposited on the substrate in a first chamber. The first piezoelectric material layer is formed on the substrate while the substrate is at a first temperature. A second piezoelectric material layer is deposited on the first piezoelectric material layer after cooling the substrate to a second temperature. The second temperature is lower than the first temperature. The first piezoelectric material layer and the second piezoelectric material layer both comprise a first piezoelectric material.

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Background/Summary

BACKGROUND

Field

(1) Examples disclosed herein relate to an apparatus and method of forming thin film layers on a substrate.

Description of the Related Art

(2) Thin film piezoelectric materials can be used in sensors and transducers. Piezoelectric sensors and transducers are also used in devices such as gyro-sensors, ink-jet printer heads and other Microelectromechanical systems (MEMS) devices, including acoustic resonator used in mobile phones and other wireless applications. These thin piezoelectric films can be fabricated by techniques such as sputtering, pulsed laser ablation (PLD), MOCVD, and sol-gel deposition.

(3) In semiconductor processing, physical vapor deposition (PVD) (e.g., sputtering process) is a conventionally used process for depositing a thin film. A PVD process includes bombarding a target that has a source material. Ions are generated in a plasma within a chamber, causing the source material to be sputtered from the target to a substrate. During some PVD processes, the sputtered source material is then accelerated towards the substrate being processed via a voltage bias. The source material is deposited on the surface of the substrate. In some examples, the

sputtered source material may react with another reactant. In the case of sputtering a fabricated layer on a substrate, epitaxial growth of the thin films can demonstrate strain and/or dislocated structure due to the thermal and lattice mismatch between the piezoelectric-based materials and the substrate.

(4) During deposition of the sputtered material, a thickness and stress uniformity of the sputtered thin films may be affected by several controlled parameters. The controlled parameters can include a strength of a magnetic field used to trap electrons near the surface of a sputtering target, a lattice match or mismatch between adjacent materials, and crystal orientation of the substrate can make it difficult to maintain uniform properties of thin films. Such non-uniformity of thin films can reduce the quality and yield in device production of piezoelectric sensors and actuators.

(5) Accordingly, there is a need for an improved method and apparatus for depositing quality piezoelectric materials and increasing the yield in producing sensing devices.

SUMMARY

(6) Disclosed herein is an apparatus and method of forming thin film layers on a substrate. A first piezoelectric material layer is deposited on the substrate in a first chamber. The first piezoelectric material layer is formed on the substrate while the substrate is at a first temperature. A second piezoelectric material layer is deposited on the first piezoelectric material layer after cooling the substrate to a second temperature. The second temperature is lower than the first temperature. The first piezoelectric material layer and the second piezoelectric material layer both comprise a first piezoelectric material.

(7) In another example, a method of processing a substrate in a cluster tool includes depositing a seed layer on a substrate in a first process chamber of the cluster tool by use of a physical vapor deposition process, and at a first temperature. The seed layer includes a first material. The substrate is cooled to a second temperature. The second temperature is below the first temperature. A first piezoelectric material layer is deposited on the seed layer by use of a physical vapor deposition process, after cooling the substrate to the second temperature. The first piezoelectric material layer includes the first material.

(8) In another implementation, an apparatus for processing a substrate is disclosed. The apparatus includes a processor coupled to at least one non-transitory computer readable medium, wherein the at least one non-transitory computer readable medium includes instructions which when executed by the processor are configured to perform a method. The method includes depositing a first piezoelectric material layer on the substrate in a first chamber. The first piezoelectric material layer is formed on the substrate while the substrate is at a first temperature. The method further includes depositing a second piezoelectric material layer on the first piezoelectric material layer after cooling the substrate to a second temperature. The second temperature is lower than the first temperature. The first piezoelectric material layer and the second piezoelectric material layer both include a first piezoelectric material.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to examples, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only examples and are therefore not to be considered limiting of its scope, and may admit to other equally effective examples.

(2) FIG. 1 is a plan view of a cluster tool that is adapted to deposit thin film layers on a substrate by use of one or more processing chambers, according to an embodiment described herein.

(3) FIG. 2 is a side cross-sectional view of one of the processing chambers that is adapted to

deposit thin film layers on the substrate in the cluster tool illustrated in FIG. 1, according to an embodiment described herein.

(4) FIG. 3 is a side cross sectional view of another processing chamber adapted to process the substrate in the cluster tool illustrated in FIG. 1, according to an embodiment described herein.

(5) FIG. 4A is a side view of an exemplary film stack produced within the cluster tool disclosed in FIG. 1, according to an embodiment described herein.

(6) FIG. 4B is a side view of an exemplary film stack produced within the cluster tool disclosed in FIG. 1, according to an embodiment described herein.

(7) FIG. 5A is a flow chart depicting an exemplary method of producing the film stack within the cluster tool illustrated in FIG. 1, according to an embodiment described herein.

(8) FIG. 5B is a flow chart depicting an exemplary method of producing the film stack **400**, according to an embodiment described herein.

(9) FIG. 6 is a flow chart illustrating a method of forming a seed layer or portion of a bulk layer, according to an embodiment described herein.

(10) FIG. 7 is a plan view of a controller that can provide instructions to the any one of the processing chambers depicted in FIGS. 1-3.

(11) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements and features of one example may be beneficially incorporated in other examples without further recitation.

DETAILED DESCRIPTION

(12) Disclosed herein is an apparatus and method of forming thin film layers on a substrate. In some embodiments, it has been found that when piezo-electric materials are deposited on a substrate at high temperatures, the variation in the crystal orientation in the deposited film is improved, resulting in narrower full-width-half-maximum (FWHM) peaks. However, under high substrate temperatures, within-wafer (WIW) stress uniformity is worse. In contrast, lower substrate temperatures correlate to better stress uniformity, but wider FWHM peaks are created (e.g., wider variation in crystalline structure). The crystalline structures of the thin film layers disclosed herein were studied using X-ray diffraction (XRD) techniques, however, other techniques may be used.

(13) Advantageously, a differential temperature control method during the formation of different parts of a deposited thin film layer can be utilized, which introduces a cooling process during part of the deposition process that enables the formation of narrower FWHM XRD peaks in the deposited film, and improved stress uniformity. In some configurations of the differential temperature control method, for example, the substrate is degassed at a high temperature in one chamber, and subsequently a seed layer is deposited on the substrate using a physical vapor deposition (PVD) process while the substrate remains at a temperature that is equal to or close to the high temperature achieved within the substrate during a prior processing step, such as a degas processing step. The substrate can then be transferred to another chamber where the substrate is cooled prior to depositing a bulk layer over the seed layer using a PVD process, at a temperature near room temperature (RT). Herein, room temperature (RT) is less than 100° C., such as from about 20° C. to about 50° C. The differential temperature control process has been found to improve the film properties of the film stack (e.g., seed layer and bulk layer) by reducing film stress, promoting a desired crystal orientation and reducing the variation in crystal orientation of the deposited film layer across the substrate surface. An example of a film stack (i.e., film stack **400**) is shown in FIG. 4.

(14) During the process of depositing one or more layers using a sputtering process (i.e., PVD process), differences in the arrangement of the atoms in the thin film layers on the substrate due to thermal properties or surface non-uniformity can result in cone defects, stacking defects, and other surface defects being formed in the deposited film layer. The defects increase surface roughness, and weaken piezoelectric coupling. An improvement in deposited thin film properties can be

achieved in various types of deposited film, such as piezoelectric materials, through the methods disclosed herein. During the process of depositing a seed layer by PVD, process parameters such as degas temperature, pressure, bias power, magnetic rotation speed, a distance between target and substrate, and gas composition ratio concentration (e.g., reactive gas (N.sub.2) to carrier gas (e.g., Ar, Kr, etc.) ratio) are adjusted in order to improve the stress, stress uniformity, crystal orientation, crystal orientation uniformity and/or thickness uniformity of the deposited thin film layer(s). By adjusting any one of or a combination of process parameters disclosed herein the number of cone- and surface-defects can also be improved in the deposited thin film layer.

(15) The methods and apparatuses disclosed herein enable fine-tuning thin film properties by adjusting various process parameters during one or more steps during the formation of the deposited film stack. Process parameters, deposition time, and a cooling recipe can be adjusted by use of a system controller during processing in order to improve stress uniformity and crystal orientation. The degas temperature, process pressure, substrate bias power, gas concentration ratio are examples of some of the adjustable process parameters. The process parameters can be adjusted alone or in combination with the deposition process time or substrate cooling recipe during or after the seed layer deposition. Accordingly, by fine tuning the above-noted parameters while depositing one or more portion of the film layer stack, an improved stress, within wafer (WIW) stress uniformity, crystal orientation, within wafer (WIW) crystal orientation uniformity and/or thickness uniformity of the deposited thin film can be achieved.

(16) FIG. 1 is a plan view of a cluster tool **100** that is adapted to deposit thin film layers on a substrate. One example of the cluster tool **100** is commercially available from Applied Materials, Inc. of Santa Clara, Calif., and is known as the Endura® system.

(17) The cluster tool **100** includes a factory interface **104**, loading dock **140**, a first robot **132**, and a second robot **136**. An orient chamber **152**, degas chamber **156**, first processing chambers **160** and **168**, second processing chambers **164-167**, a second robot **136**, and a main frame **172** are included in the cluster tool **100**. The cluster tool **100** also includes a first transfer chamber **124** and a second transfer chamber **128**.

(18) Each cassette **112**, or FOUP, is configured to receive a plurality of substrates **201**. In this configuration, the substrates **201** are removed from the cassette **112**, by one of the factory interface robots **120**. The factory interface robots **120** will transfer the substrate **201** from the cassette **112** and load the substrate **201** into the loading dock **140** (i.e., load lock). Upon completion of substrate processing in the cluster tool **100**, the processed substrates **201** are then returned to their respective cassette **112**.

(19) A main frame **172** includes the first transfer chamber **124** that includes a first robot **132**. The first robot **132** is configured to move the substrate **201** between the orient chamber **152**, degas chamber **156**, and a first processing chamber **160**. Each of the orient chamber **152**, degas chamber **156**, and a first processing chamber **160** is arranged around the periphery of the first transfer chamber **124**. In some configurations, the first transfer chamber **124** is vacuum pumped to a moderately low pressure, for example, about 1 milliTorr or less. The second transfer chamber **128** is pumped to a lower pressure, for example, 1 microTorr or less. Accordingly, the first transfer chamber **124** or the second transfer chamber **128** is maintained at least at a moderate vacuum level to prevent the transfer of contamination between the first transfer chamber **124** and the second transfer chamber **128**. It is understood that any discussion or description of the first processing chamber **160** necessarily includes the first processing chamber **168**, unless explicitly stated otherwise.

(20) A second robot **136** is configured to move the substrate(s) **201** between the first processing chamber **160** and a second processing chamber **164**. The second robot **136** is disposed within the second transfer chamber **128**. The second robot **136** is configured to transfer substrates **201** to and from the first processing chamber **160** and the second processing chamber **164** or other process chambers **165-168** attached to the second transfer chamber **128** portion of the main frame **172**. In

one configuration, each of the first robot **132** and the second robot **136** is a “frog-leg” type robot, available from Applied Materials, Inc. of Santa Clara, Calif. The first transfer chamber **124** can be selectively isolated from each of the orient chamber **152**, degas chamber **156**, and first processing chamber **160** by use of slit valves (not shown) that are disposed between each of the orient chamber **152**, degas chamber **156**, and first processing chamber **160**. The second transfer chamber **128** can be selectively isolated from each of the first processing chamber **160** and the second processing chamber **164** by use of slit valves that are disposed between each of the first processing chamber **160** or the second processing chamber **164**. Herein, it is understood that any discussion or description of the second processing chamber **164** includes any one of the second processing chambers **165-167**.

(21) Each loading dock **140** is each selectively isolated from both the first transfer chamber **124** by slit valves and from the exterior region **116** of the factory interface **104** by vacuum doors (not shown). In this configuration, the factory interface robots **120** in the factory interface are configured to move a substrate **201** from a cassette **112** to the loading dock **140**. The substrate **201** is then isolated from the exterior region **116** of the factory interface **104** by the vacuum door (not shown) coupled to the loading dock **140**. The substrate **201** is transferred to the loading dock **140**. After the loading dock **140** is pumped down to a desired pressure, the substrate **201** can then be accessed by the first robot **132** through a slit valve opening (not shown) formed between the first transfer chamber **124** and the loading dock **140**.

(22) Each substrate **201** is loaded into a cassette **112** that is coupled to a factory interface **104**. The substrate **201** may have a diameter in a range from about 100 mm to about 750 mm. The substrate **201** may be formed from a variety of materials, including Si, SiC or SiC-coated graphite. In one example, the substrate **201** includes a silicon carbide material and has a surface area of about 1,000 cm.² or more. In another example, the surface area of the substrate **201** may be about 2,000 cm.² or more, and about 4,000 cm.² or more.

(23) One or more orient chambers **152** may be used to align the substrate **201** in a desired rotational orientation within the cluster tool **100**. By aligning the substrate **201**, the substrate **201** is also aligned. The orient chamber **152** may be positioned proximate the loading dock **140** and proximate the degas chamber **156**.

(24) In some embodiments, the orient chambers **152** include a heat source, such as lamps or infrared generating radiant heaters. The heat source within the orient chambers **152** can be adapted to heat the substrate **201** and each substrate **201** to a desired temperature. The orient chambers **152** can be pressurized under a vacuum condition to ensure that any undesirable water or other contamination is removed from the surface of the substrate **201** prior to processing in other downstream chambers.

(25) In some embodiments, the cluster tool **100** includes a pre-clean chamber **156** that is adapted to clean the surface of a substrate **201** by use of cleaning process that includes exposing the surface of the substrate to a radio frequency (RF) generated plasma and/or one or more pre-cleaning gas compositions that includes a carrier gas (e.g., Ar, He, Kr) and/or a reactive gas (e.g., hydrogen). In some embodiments, the pre-clean chamber **156** is adapted to perform a process that may include a non-selective sputter etching process. The pre-clean chamber **156** will typically include components similar to the components found in the processing chamber **300**, which is described below in conjunction with FIG. 3.

(26) Each first processing chamber **160** is configured to process the substrate(s) **201** therein. Processing may include cooling the substrate, heating the substrate **201**, etching and/or depositing one or more layers on a surface of the substrate **201**. In one configuration, the first processing chamber **160** is configured to cool or heat a substrate **201**.

(27) Each of the processing chambers **164-167** are adapted to perform an etch and/or deposition process. In some embodiments, the deposition process may include a sputter deposition process (i.e., PVD deposition process). The sputter deposition process may also include temperature

regulation step that is adapted to a cool down and/or control the temperature of the substrate during processing.

(28) FIG. 2 is a plan view of a process chamber **200** that can be part of one or more of the processing chamber **164-167**. The process chamber **200** is adapted to deposit thin film layers on the substrate **201** in the cluster tool illustrated in FIG. 1. The processing chamber **200** can be a magnetron type PVD chamber available from Applied Materials, Inc. of Santa Clara, Calif. The processing chamber **200** includes a chamber **204**, a target **248**, a magnetron **296**, a vacuum pumping system **260**, a substrate support assembly **232**, and a process kit **214**. In one example, the target **248** is a Scandium (Sc) doped aluminum (Al) target. In another example, the target **248** may be made from Al.

(29) The chamber **204** supports the target **248**, which is sealed at one end of the chamber **204** through a target isolator **244** using a plurality of O-rings. The chamber **204** may be held under vacuum.

(30) The process kit **214** includes an edge ring **234**, a first shield **216** and the second shield **220** that are separated by a second dielectric shield isolator **224**. The process kit **214** parts are positioned within the chamber **204** to protect the chamber wall **208**. A metal within the process kit **214** is electrically grounded from the sputtered material that is generated in an interior volume **212**. The first shield **216** may be permitted to float electrically and the second shield **220** is electrically grounded. In an alternate example, either or both of the first shield **216** or second shield **220** may be grounded, floating or biased to the same or different non-ground levels. The first shield **216** and second shield **220** may be made of stainless steel. An inner surface **228** may be bead-blasted or otherwise roughened to promote adhesion of the material sputter deposited on the inner surface **228**.

(31) The substrate support assembly **232** includes a pedestal **236**. The pedestal **236** may include an electrostatic chuck **238** that has a supporting surface that is adapted to support a substrate **201** over an electrode **240**. It is appreciated that other devices may be used to hold the substrate **201** in place during processing. Resistive heaters, refrigerant channels, and/or thermal transfer gas cavities, which are not illustrated in FIG. 2, may be formed in the pedestal **236** to provide thermal control of the substrate **201** during processing. The electrode **240**, which is coupled to a first power supply **252**, may apply an RF and/or a DC bias to the substrate **201** to attract a plasma **299** ionized deposition material and processes gases. In other applications, biasing of the substrate **201** may be reduced or eliminated to further reduce the potential for damage to the deposited thin film layer.

(32) The target **248** has at least a surface portion made of a material to be sputter deposited on substrate **201**. In one example, a pulsed DC, RF and/or a pulsed RF bias signal is applied to the target **248** by a second power source **284**. The pulsed DC, RF and/or pulsed RF bias signal enables the deposition of an optional non-conductive layer, such as a PZT or aluminum nitride layer. In order to attract the ions generated by the plasma **299** to sputter the target **248**, the target **248** may be biased by the second power source **284** to provide an average power of 1 to 20 KW, for example. The pulsed DC and/or RF bias signal applied to the target **248** may include a signal that has a plurality of alternating first and second intervals (detailed below). During each of the first intervals, the voltage of the applied bias signal is negative to attract ions to sputter the target **248**. During the alternating second interval, the applied bias signal is lower than the bias applied during the first interval, unbiased (e.g., zero applied voltage). In some examples, the applied bias signal has a positive voltage to repel positively charged ions from the target **248** to reduce arcing.

(33) One skilled in the art will appreciate that the pulsed bias signal applied to the target **248** can provide many beneficial processing advantages, depending upon the particular application. For example, the pulsed bias signal may be used to reduce the deposition rate, form plasma **299**, and increase the peak energy in the plasma **299** for effectively controlling a plasma chemistry to form a film stack **400** (illustrated in FIG. 4). For example, thin film layers which are closer to stoichiometric proportions might be obtained when a pulsed biasing signal is applied to the target

248. Still other possible features include an increase in thin film quality, particularly for the film stack **400**. Additionally, thin film sheet resistance may be reduced, due to possible elimination of undesirable micro voids and columnar structures, when a pulsed bias is applied. It is appreciated that in some examples, a non-pulsed biasing signal may be applied to bias the target **248** during one or more parts of the deposition process, or in combination with a pulsed bias signal. The non-pulsed biasing signal can be a constant DC or an RF power level bias signal.

(34) The substrate **201** mounted on the pedestal **236** can be biased to attract or repel ions generated in the formed plasma **299**. For example, the first power supply **252** may be provided to apply RF power to the pedestal **236** to bias the substrate **201** to attract deposition material ions during the deposition process. In addition, the first power supply **252** may be configured to apply RF power to the electrode **240** of pedestal **236** to couple supplemental energy to the plasma **299**. During the deposition process, the pedestal **236** may be electrically floating. Accordingly, a negative DC bias may nonetheless develop on the pedestal **236**. Alternatively, the pedestal **236** may be biased by a source at a voltage of between -1000 Volts to +500 Volts, such as about -30 VDC. For example, the pedestal **236** may be biased in order to bias the substrate **201**, attracting the ionized deposition material to the substrate **201**. In some configurations, a capacitor tuner (not shown) can be used with the second power source **284** to control the floating potential on the substrate **201** during processing. In an alternative example, the substrate **201** may be left floating electrically.

(35) If the first power supply **252**, used to bias the substrate **201** through the pedestal **236**, is an RF power supply, the supply may operate at a frequency of about 13.56 MHz to about 60 MHz. The pedestal **236** may be supplied with RF power in a range of 10 watts to 5 kW. A computer-based controller **256** may be programmed to control the power levels, voltages, currents and frequencies. Accordingly, it is understood that the above-mentioned power level, voltage level, and frequencies may vary according to the program.

(36) The vacuum pumping system **260** includes a pump assembly **268** and valve **264**. The pump assembly **268** may include a cryopump, roughing pump(s) (not shown) that are used to maintain a desirable pressure in the interior volume **212** of the processing chamber **200**.

(37) The magnetron **296** is disposed adjacent to and is rotated relative to the target **248**. A plurality of magnets **298** is included in the magnetron **296**. The plurality of magnets **298** includes plural polarized magnets N, and plural magnets S having an opposite polarization to magnets N. The magnets **298** are used to confine plasma **299** generated in the interior volume **212** by biasing the target **248** using the second power source **284** to sputter material from a front surface **250** of the target **248**. The second power source **284** has a second power supply **288** that is configured to deliver DC and/or RF power to the target **248**. In some example, delivery of RF power to the second power source **284** may also include a match circuit **290**.

(38) The magnetron **296** can be tilted with respect to a surface of the target **248**, such as the front surface **250** or back surface **251**. In other words, the magnetron **296** forms an angle **203** with respect to a central axis **202** or the axis of rotation of the magnetron **296**. The tilting of the magnetron **296** at an angle **203** may be controlled by the controller **256** via the motor **292**. The degree of the tilting of the magnetron **296** may be adjusted between processing batches, between substrates **201**, or in-situ during the processing of the substrate(s) **201**. The angle **203** at which the magnetron **296** is tilted may be controlled based on thin film thickness or stress data feedback. The specific component of the magnetron **296** that is tilted at the angle **203** with respect to the target **248** may vary. In one example, the longitudinal dimension of the backing plate **293** (e.g., yoke) is tilted at the angle **203** with respect to the target **248**. In one example, a plane **294** (e.g., parallel to the X-Y plane) defined by lower ends of magnets **298** facing the back surface **251** of the target **248** is tilted at the angle **203** with respect to the target **248**. In one example, the magnetron **296** is tilted with respect to the back surface **251** of the target **248**. In another example, the magnetron is tilted with respect to the front surface **250**.

(39) In one embodiment, when the magnetron **296** is rotated about the central axis **202** by a rotation

motor **292** during processing, the angle **203** is maintained between the magnetron **296** and the target **248**, such that any point on the magnetron **296**, as is rotated about the central axis **202**, will remain the same vertical distance (i.e., Z-direction distance) from a surface of the target **248**, such as the back surface **251**. As the magnetron **296** rotates, the strength of the magnetic field produced by the magnetron **296** is an average of the various strengths of magnetic fields produced by each magnet **298**. The magnetic field is averaged across the front surface **250** of the target **248**. The averaging of the strengths of the magnetic fields enables uniform thin film properties and uniform erosion of the target **248**.

(40) The angle **203** is determined by establishing an angle between the back surface **251**, as shown in FIG. 2, of the target **248** and the plane **294** of the magnetron **296**. Another manner of determining the angle **203** is establishing an angle between the front surface **250** of the target **248** and the plane **294** of the magnetron **296**. For simplicity, any discussion of the angle **203** formed between the magnetron **296** and the target **248**, necessarily include the surfaces **250** or **251** of the target **248** and the plane **294** of the magnetron **296**. The angle **203** formed between the magnetron **296** and the target **248** may be from about 0.2 degrees to about 5 degrees. In another example, the angle **203** is from about 1 degree to about 2 degrees. If the angle **203** of the tilting of the magnetron **296** with respect to the target **248** is less than about 0.2 degrees, the effect of the averaging of the magnetic fields strengths can be diminished. As noted, the angle **203** of the tilted magnetron **296** with respect to the target **248** is less than or equal to about 5 degrees. Accordingly, the magnetron **296** and the central axis **202** form an acute angle (not labelled) ranging from about 85 degrees to about 89.7 degrees. The acute angle may range from about 88 degrees to about 89 degrees. A sum of the angle **203** and the acute angle is typically 90 degrees.

(41) A first gas source **272** supplies a gas to the chamber **204** through a mass flow controller **280**. One example of the gas is a chemically inactive noble gas, such as argon (Ar). The gas can be admitted to the top of the chamber **204**, or as illustrated, at the bottom of the chamber **204**. One or more inlet pipes (not illustrated) penetrate apertures through the bottom of a second shield **220**. Alternatively, inlet pipes may be coupled to apertures within the pedestal **236**. During PVD processes, a nitrogen (N) gas may be delivered from a second gas source **276** to form a layer on the substrate **201**. The layer may include a material such as aluminum nitride (AlN).

(42) FIG. 3 is a cross sectional view of another processing chamber **300** adapted to process the substrate in the cluster tool illustrated in FIG. 1. An example of the processing chamber **300** useful for the present disclosure is the Pre-Clean II Chamber available from Applied Materials, Inc., Santa Clara, Calif.

(43) The pre-clean chamber **300** has a substrate support assembly **304** disposed in a chamber enclosure **308** under a dome **312**. In one example, the dome **312** may be made from quartz. The pedestal **320** includes the substrate support assembly **304** having a substrate support **316**. The substrate support **316** is disposed within a recess **324** on the substrate support assembly **304**. During processing, the substrate **201** is placed on the substrate support **316**. At least one locating pin **328** retains the substrate in a desired lateral position on the substrate support **316**.

(44) A coil **344** is disposed outside of the dome **312** and connected to an RF power source **332**. The RF power source **332** initiates and maintains a plasma formed from the process gases within the processing chamber **300**. An RF match network **340** is provided to match the RF power source **332** and the coil **344**. The substrate support assembly **304** is connected to a DC power source **336** that provides a bias to the substrate support assembly **304**.

(45) The substrate **201** may be pre-cleaned or etched using the plasma in the processing chamber **300** prior to depositing one or more layer within a film stack **400**, such as seed layer **404** (shown in FIGS. 4A and 4B). Once the substrate **201** is positioned for processing in the processing chamber **300**, a processing gas is introduced into the interior volume **212**. The processing gas may include between about 0% and about 100% hydrogen (H) and the balance a carrier gas. The processing gas may be between about 5% and about 50% of H.

(46) The processing gas can include a carrier gas, such as Ar or helium (He), at a concentration of between about 95% and about 50%. The processing gas is ignited in the interior volume **212** to form the plasma, thus subjecting the substrate **201** to the plasma. For example, plasma may be generated by applying between about 50 W and about 500 W of power from the RF power source **332** to the coil **344**. The DC power source **336** may also provide power between about 10 W and about 300 W of DC bias power. The plasma may be maintained for a period between about 10 seconds and about 300 seconds. Once the pre-cleaning process is completed, processing chamber **300** is evacuated to exhaust the processing gas and the reacted byproducts from the processing chamber **300**.

(47) FIG. **4A** is a side view of an exemplary film stack **400** produced within the cluster tool disclosed in FIG. **1**. The film stack **400** includes the substrate **201** and the seed layer **404**, and may also include the bulk layer **408**. A seed layer **404** is formed on a surface of the substrate **201**. A bulk layer **408** is formed on top of and in contact with a surface of the seed layer **404**. In at least one example, a material of the seed layer **404**, the bulk layer **408** are the same material, and the physical properties of the seed layer **404** and bulk layer **408** are different. In another example, the seed layer **404** and the bulk layer **408** may be formed from a different material. Alternatively, as shown in FIG. **4B**, the bulk layer **408** can be formed from several interlayers **412**.

(48) FIG. **4B** is a side view of the exemplary film stack **400** shown in FIG. **4A** illustrating an alternative formation of the bulk layer **408**. The film stack **400** may include the substrate **201**, the seed layer **404**, and the bulk layer **408**. A seed layer **404** is formed on a surface of the substrate **201**. A bulk layer **408** is formed on top of and in contact with a surface of the seed layer **404**. The bulk layer **408** may include one or more interlayers **412**, where the interlayers **412** may be expressed as **412(n)**, where n is the number of interlayers **412** in the bulk layer **408** that may vary from 1 to n . For example, while $n=4$ in the example illustrated in FIG. **4**, in an alternate example, n is from 1 to 30. As stated above, in at least one example, a material of the seed layer **404**, and one or more of the interlayers **412** of the bulk layer **408** are formed of the same material. In this example, at least some of the physical properties of the seed layer **404**, and the one or more interlayers **412** are different.

(49) In one example, the seed layer **404** and bulk layer **408** are made from a piezoelectric material such as scandium-doped aluminum nitride (ScAlN) or aluminum nitride (AlN). In another example, the piezoelectric material can be any one of or a combination of titanium nitride (TiN), hafnium nitride (HfN), or silicon nitride (SixN.sub.y). The substrate **201**, in some examples, has a crystal orientation of $\langle 001 \rangle$. The substrate **201** may be made from other metals having an appropriate lattice, including but not limited to a polycrystalline molybdenum, and AlN.

(50) FIG. **5A** is a flow chart depicting an exemplary method **500** of producing the film stack **400** within the cluster tool illustrated in FIG. **1**.

(51) At block **502**, the substrate **201** is loaded into the loading dock **140** of the cluster tool **100**. In an example, the first robot **132** moves the substrate **201** to the orient chamber **152**. The substrate **201** is passed through the first transfer chamber **124** by the first robot **132**. The robot **132** in the first transfer chamber **124** moves the substrate **201** from the orient chamber **152** to the first processing chamber **160**. As stated above, pressure (P) in the first transfer chamber **124** may be about 1 microTorr. Accordingly, the pressure is held in a vacuum state.

(52) At block **504**, the substrate **201** is disposed within a degas chamber **156** or a degas process is performed in the orient chamber **152**. In one embodiment, the substrate **201** is transferred from the loading dock **140** into the degas chamber **156**. The substrate **201** is moved through the first transfer chamber **124** as the substrate **201** moves from the loading dock **140** to the degas chamber **156**. Alternatively, the first robot **132** transfers the substrate **201** from the loading dock **140** to the orient chamber **152**. The first robot **132** transfers the substrate **201** from the orient chamber **152** to the first processing chamber **160**. The temperature (T) of the substrate **201** can be any temperature between about 120 degrees Celsius and about 600 degrees Celsius, such as about 200° C. to about

600° C. In another example, temperature of the substrate **201** can be any temperature between about 140° C. and about 500° C.

(53) Optionally, a surface of the substrate **201** is exposed to a plasma at block **506** to preclean the surface of the substrate. In some embodiments, the processes performed in block **506** are performed after performing the processes in block **504**. Alternatively, in some embodiments of the method **500**, the processes performed in block **506** are performed before performing the processes in block **504**, as illustrated in FIG. 5B. While FIG. 5B includes the positions of blocks **504** and **506** in method **500** being in a different order than the order shown in FIG. 5A, this configuration is not intended to limit the method illustrated in FIG. 5B to that specific order of processes. During block **506** the surface of the substrate can be pre-cleaned (e.g., bombarded with reactive gas (e.g., H.sub.2) or non-reactive gas (e.g., Ar, Ne, He) ions and/or gas radicals (e.g., etched)) in the processing chamber **300**. The processes performed during block **506** can also include heating, maintaining the incoming temperature of the substrate or cooling the substrate **201**. Pre-cleaning the surface of the substrate **201** prior to depositing the seed layer **404**, can reduce surface defects in the seed layer **404** when the seed layer **404** is deposited on the substrate **201**.

(54) The seed layer **404** is formed on the substrate **201** at block **508** using the second processing chamber **164**. The second processing chamber **164** receives the substrate **201** from the first processing chamber **160**. The substrate **201** passes through the second transfer chamber **128** as the substrate **201** is moved from the first processing chamber **160** to the second processing chamber **164**. The seed layer **404** can be formed in the second processing chamber **164** that includes the components shown in processing chamber **200**. The process performed during block may include delivering a pulsed DC, RF and/or a pulsed RF bias signal to the target **248** by the second power source **284**. In order to attract the ions generated by the plasma **299** to sputter the target **248**, the target **248** may be biased by the second power source **284** to provide an average power of 1 to 20 KW, for example. The substrate **201**, during block **508**, can be mounted on or positioned over the pedestal **236** so that the temperature of the substrate can remain at, or can be substantially maintained at, the incoming temperature of wafer (e.g., degas process temperature or pre-clean process temperature, etc.). In one example, the incoming substrate temperature, and thus substrate processing temperature, is greater than room temperature, such as between about 120° C. and about 600° C. The substrate **201** can additionally be mounted on the pedestal **236** that can be biased. For example, the first power supply **252** may be provided to apply RF power to the electrode **240** in the pedestal **236** to bias the substrate **201** to attract deposition material ions during the deposition process, such as biasing the substrate at a voltage of between -1000 Volts to +500 Volts, such as about -30 VDC. The seed layer **404** may have a thickness from about 10 nm to about 100 nm. In one example, the seed layer **404** has a thickness from about 10 nm to about 50 nm. It has been found that, by maintaining a high initial temperature (e.g., degas temperature), defects formed in the seed layer **404** decrease. It has been found that, defects in the seed layer **404** further decreases as a bias power, applied to the substrate **201**, is increased. Therefore, in some embodiments a high initial temperature (e.g., degas temperature) is maintained on the substrate **201** while a bias power is applied to the substrate **201**. In one example, the incoming substrate **201** temperature, and thus substrate processing temperature, is between about 120° C. and 600° C. and a substrate bias of about -30 VDC is applied to the substrate **201** by biasing the electrode **240** in the substrate support (i.e., pedestal **236**) during processing. The second robot **136** moves the substrate **201** from the second processing chamber **164** to the first processing chamber **160**.

(55) FIG. 6 is a flow chart of a process of forming a seed layer **404** by use of a seed layer formation block **508**, which is illustrated in FIG. 5A or 5B. It is understood that any one of or, all of blocks **600-612** may be carried out during performance of the seed layer **404** formation process performed during block **508**.

(56) At block **600**, a bias power applied to the electrode **240** can be adjusted during the deposition process completed during block **508**. In this example, a bias power may be either one of a first bias

power P1 or a second bias power P2. The bias power may be between about 10 Watts to about 200 Watts. As the bias power increases, the defects formed within the seed layer **404** have been found to decrease. Thus, in some embodiments, a bias power greater than about 50 W, or greater than 100 W, or even greater than 120 W is applied to the electrode **240** during the deposition process completed during block **508**.

(57) At block **602**, a pressure at which the substrate **201** is subjected to during the formation of the seed layer **404** at block **508** can be reduced. The pressure may be reduced to about 15 mTorr or less. By reducing the pressure at which the seed layer **404** is formed, a reduction in defects in the seed layer **404** can be achieved.

(58) A magnetic rotation speed of the magnetron **296** is increased at block **606**. The magnetic rotation speed may be between about 30 rotations per minute (rpm) and about 120 rpm. In one example, the magnetic rotation speed may be between 50 rpm include speeds up to about and about 90 rpm. In some configurations, the magnetic rotation speed can vary between 90 rpm and about 120 rpm. Surface defects on the seed layer **404** may be reduced by increasing the rotation speed of the magnetron **296** during the formation of the seed layer **404** during block **508** versus the rotation speed that is used to form the bulk layer **408** during block **512**. The magnetic rotation speed of the magnetron **296** can be also be adjusted with during the formation of the seed layer **404**.

(59) At block **608**, a distance from the target **248** to the surface of the substrate **201** is adjusted during the formation of the seed layer **404** during block **508**. Otherwise stated, the distance between the surface of target **248** and the surface of the substrate **201** is established. The distance between the target **248** and substrate **201** can be from about 30 mm to about 70 mm. As the distance increases from 30 mm to about 70 mm, uniformity of the seed layer **404** increases. Accordingly, surface defects on the seed layer **404** are reduced.

(60) At block **610**, a duty cycle of the signal produced by the second power source **284** can be adjusted during the formation of the seed layer **404** during block **508**. As articulated above, the signal may be a DC, RF and/or pulsed RF bias signal. The seed layer **404** may be deposited at a seed duty cycle D.sub.seed. The seed duty cycle D.sub.seed may be from 60% to 90%.

Accordingly, the second power source **284** can be turned “on” from about 60% to 90% of the duty cycle. Correspondingly, the second power source **284** may be “off” for a period of 10% to 40% of the duty cycle. In one example, the power “on” time is 88.5% and “off” time is 12.5% of the duty cycle. In another example, the power “on” time is 67% and “off” time is 33%. By reducing the time in which the signal is “off,” seed layer **404** surface defects are reduced. In some examples, the bulk layer **408** can be deposited at a different duty cycle.

(61) At block **612**, a concentration of nitrogen (N) in the processing gas provided to the chamber **204** is adjusted during the formation of the seed layer **404** during block **508**. In this example, the concentration of N is between 100% of the processing gas and 66% of the processing gas. As the concentration of N in the processing gas increases, the surface defects on the seed layer **404** are reduced. In some examples, the nitrogen contribution is N.sub.2. As stated above, each of the blocks **600-612** can be adjusted individually or in combination with the other blocks during the formation of the seed layer **404**, and/or during the seed layer formation block **508**.

(62) Referring back to FIG. 5A, at block **510**, the substrate **201** is cooled in the first processing chamber **160** to a temperature less than 100° C., such as a temperature between about room temperature (e.g., ~20° C.) and 50° C. During block **510**, the substrate **201** can be cooled in an environment that includes Ar and N.sub.2. A ratio of Ar to N.sub.2 may be about 5 to 1. In an exemplary configuration, the ratio of Ar to N may be about 3 to 1. The pressure (P) may be from about 3 Torr to about 15 Torr, during the process performed during block **510** to rapidly cool the substrate **201**. The substrate **201** is cooled until the substrate **201** reaches a designated temperature (T). In some embodiments, the substrate **201** is cooled in the same chamber where the seed layer **404** is deposited. Alternatively, the substrate **201** can be cooled in a different chamber than where the seed layer **404** is deposited. In one non-limiting example, it may take from about 1 second to

about 100 seconds in order for the substrate **201** to reach the designated temperature. In one configuration, the substrate **201** may reach the designated temperature in about 20 second to about 50 seconds. The pressure (P) at which the substrate **201** is cooled at block **510** is higher than the pressure at which the substrate **201** is processed at block **502** to block **508**. In some configurations, as noted above, it is desirable to perform the processes performed in block **510** in a separate process chamber from the chamber the seed layer is formed in due to the time that would be required to remove the amount of gas needed to achieve the processing pressure achieved during block **510** from the processing chamber the seed layer is formed in before forming another seed layer on another substrate. The temperature may be reduced at a rate from about 600° C. to about 200° C. in about 1 second to about 100 seconds, such as about between 1 second and 20 seconds. In an alternate example, the temperature may be reduced at a rate from about 600° C. to about 200° C. in about 20 seconds to about 50 seconds. In an example where the substrate **201** is cooled in the different chamber than where the seed layer **404** is deposited, the temperature of the substrate **201** may be reduced to at a rate between about 50° C. and about 100° C. in about 1 second to about 100 seconds. Alternatively, the substrate **201** may be cooled in from about 20 seconds to about 50 seconds. In yet another example, the temperature may be reduced at a rate from about 600° C. to about 250° C. in 1 second to about 100 seconds, such as about 20 seconds to about 50 seconds.

(63) At block **512**, the bulk layer **408** is formed on the substrate **201**. The bulk layer **408** can be formed in the second processing chamber **164** that includes the components shown in processing chamber **200**. A bulk layer **408** includes at least one interlayer **412** that is formed on the substrate **201** in the second processing chamber **164**. In one example, the bulk layer **408** can be formed to have a thickness from about 1 micron to 10 micron. However, the thickness of the bulk layer **408** is not limited to this range and may be deposited on the substrate **201** to any desired thickness. Because the bulk layer **408** is deposited near room temperature (RT), a temperature at which the bulk layer **408** is deposited is lower than the temperature at which the seed layer **404** is deposited. As stated above, room temperature is less than about 100° C.

(64) FIG. 5B is a flow chart depicting a version of the exemplary method **500** of producing the film stack **400** that includes a process for forming the seed layer **404** or the bulk layer **408** by performing a series of sequential processing steps, according to another embodiment of the disclosure. Block **508** and block **512** in FIG. 5B are substantially the same as described with respect to FIG. 5A with the exception that the intermediate steps shown in blocks **509A** to **509C** of block **508** and the intermediate steps shown in blocks **513A** to **513C** of block **512** have been added. Additional blocks **509A** to **509C** performed at block **508**, and additional blocks **513A** to **513C** performed at block **512** are disclosed below and utilize substantially the same process sequence steps that may form film layers that either have similar or differing material properties. In at least one example, where differing material properties of the seed layer **404** and bulk layer **408** is desired, include cases where the seed layer **404** and the bulk layer **408** have different film thicknesses and/or utilize different process settings to achieve different material properties (e.g., film stress, crystal orientation, electrical properties (e.g., resistivity)) during the performance of one or more of the blocks **509A**-**509C** versus the performance of one or more of the blocks **513A**-**513C**. The processes performed during blocks **509A**-**509C** may include the use of one or more of the processes or process settings described in conjunction with FIG. 6 to form at least a portion of the seed layer **404**.

(65) In some embodiments, the seed layer **404** is formed at block **508** on the substrate **201**, and block **508** may include two or more processing steps, such as blocks **509A** to **509C** as shown in FIG. 5B. In one embodiment, block **509A** includes the formation of a first interlayer **412(1)** that is deposited at a temperature that is significantly higher than the temperature that the subsequent interlayers (e.g., interlayers **412(2)**-**412(n)**) of seed layer **404** are formed. In some embodiments, the first interlayer **412(1)** of the seed layer **404** is formed at a temperature between about 120° C. and 600° C. In some embodiments, the temperature of a substrate disposed in a processing region

of the processing chamber **164** during formation of interlayers **412(1)-412(n)** is substantially constant. The seed layer **404** is formed in an environment of Ar and N. The ratio of Ar to N may be about 2 to 1. During block **509A** the first power supply **252** applies a first bias power **P1** to the substrate **201** using the electrode **240** for a first duration of time (**t1**). A first interlayer **412(1)** is formed on the film stack **400**. The first bias power **P1** is between about 30 Watts and about 150 Watts.

(66) A second interlayer **412(2)** is formed on top of and in contact with the first interlayer **412(1)**, at block **509B**. A second power bias **P2** is applied to the substrate **201** through the electrode **240** from the first power supply **252**, for a second duration of time (**t2**). The first duration of time is between about 1 second and about 200 seconds. The second duration of time is greater than 250 seconds and less than or equal to about 2000 seconds. In one example, the first time may be about 470 seconds and the second time can be about 1080 seconds. In some embodiments, the second interlayer **412(2)** of the seed layer **404** is formed at a temperature that is lower than or equal to the temperature that first interlayer **412(1)** is formed, and can be in a temperature range between about 120° C. and 600° C. A second bias power **P2** is less than about 120 Watts and greater than or equal to 20 watts. For example, the first bias power **P1** may be about 100 watts and the second bias power **P2** may be about 60 watt. In another example, the second bias power **P2** can be about 80 Watts.

(67) The first interlayer **412(1)** and the second interlayer **412(2)** can be formed in the same second processing chamber **164**. In an alternative example, the first interlayer **412(1)** and the second interlayer **412(2)** may be formed in a different processing chamber of the cluster tool **100**, such as one of processing chambers **165-167**. Accordingly, in at least one example, the first bias power **P1** is higher than the second bias power **P2**. In the same example, the second duration of time (**t2**) is greater than the first duration of time (**t1**). An exemplary frequency of either the first bias power **P1** or second bias power **P2** is about 13.56 MHz.

(68) For simplicity of discussion, the term interlayer **412** is used to describe the formation of the multiple layers of the seed layers **404** and multiple layers of the bulk layer **408**. However, while the same numbering for the interlayer **412** is used to describe the formation of the seed layer **404** and bulk layer **408**, each seed layer **404** and bulk layer **408** are different structures of the film stack **400**. As noted above, the bulk layer **408** can be deposited on the substrate **201**, at block **513A** to block **513B**, in substantially the same manner as the seed layer **404** is formed on the substrate **201** at blocks **509A** to **509C**.

(69) At block **509C**, the temperature of the film stack **400** is reduced. As previously stated, the film stack **400** includes the substrate **201**, the seed layer **404**, and the bulk layer **408**. In one example, the film stack **400** is transferred from the second processing chamber **164** to the first processing chamber **160**. The first processing chamber **160** may be maintained at a reduced temperature, when the substrate **201** is cooled in the first processing chamber **160**. In another example, the film stack **400** is not transferred to the first processing chamber **160**, but is subject to the reduced temperature in the second processing chamber **164**. The reduced temperature may be achieved by removing heat from the interior volume **212**, e.g., by discontinuing the supply of power to the second power source **284** or chucking the substrate **201** to the electrostatic chuck **238**. In some configurations, the film stack **400** may be actively cooled at block **509C**. When the film stack **400** is actively cooled, the substrate **201** of the film stack **400** is placed on top of and in direct contact with a temperature regulated body, such as the substrate support **316**. Alternatively, the film substrate **201** may be placed on top of and in direct contact with the electrostatic chuck **238**. In this configuration, the substrate support **316** or the electrostatic chuck **238** have a temperature lower than the substrate **201**, and thus can be used to actively cool the substrate **201**. The temperature is reduced for a period time, $\Delta t_{\text{sub.cool}}$. The period of time $\Delta t_{\text{sub.cool}}$ can be any period of time from about 1 second to about 50 seconds. In one example, the period of time $\Delta t_{\text{sub.cool}}$ is between about 10 seconds and 20 seconds. In another example, the period of time $\Delta t_{\text{sub.cool}}$ is about 40 seconds. As

the period of time Δt .sub.cool increases, stress (MPa) decreases between the crystal structures in the film stack **400**. As stated above, the substrate **201**, and therefore the film stack **400**, can be cooled in an environment that includes Ar and N.sub.2.

(70) Once the seed layer **404** of the film stack **400** reaches a desired thickness by performing one or more of the blocks **509A-509C** multiple times, according to the method **500** disclosed herein, the process will continue onto blocks **510** and **512**. As illustrated in FIG. 5B, the processes performed in blocks **513A-513C**, which are shown within block **512**, may be performed sequentially one or more times to form the bulk layer **408**. As noted above, the processes performed during blocks **513A-513C** will include depositing two or more layers and cooling the substrate using processes similar to the processes described in blocks **509A-509C**. Also, as noted above, the two or more film layers formed and processed during the performance of blocks **513A-513C** can include using differing process parameters to achieve multiple film layers that have different film thicknesses and/or utilize different process settings to achieve different material properties (e.g., film stress, crystal orientation, electrical properties (e.g., resistivity)) than the two or more layers formed during the performance of blocks **509A-509C**. The two or more film layers formed and processed during the performance of blocks **513A-513C** may each also have different film thicknesses and/or different material properties.

(71) After blocks **502-512** have been completed, and the film stack **400** is formed, the substrate is then returned to the factory interface **104** by use of a process in which the substrate and film stack **400** is transferred from the first processing chamber **160** to the loading dock **140** by the first robot **132**.

(72) FIG. 7 is a plan view of a controller **700** that can provide instructions to the any one of the processing chambers depicted in FIGS. 1-3.

(73) An optional display unit **701** may be coupled to the controller **700**. The controller **700** includes a processor **704**, a memory **708**, and support circuits **712** that are coupled to one another. The controller **700** may be on-board the cluster tool **100**, or in an alternative example, the controller **700** may be on-board one of the processing chambers in FIG. 2 or 3, or a remote device (not shown).

(74) The display unit **701** includes an input control unit, such as power supplies, clocks, cache, input/output (I/O) circuits, coupled to the various components of the display unit **701** to facilitate control thereof. The processor **704** may be one of any form of general purpose microprocessor, or a general purpose central processing unit (CPU), each of which can be used in an industrial setting, such as a programmable logic controller (PLC).

(75) The memory **708** includes at least one non-transitory computer readable medium and may be one or more of readily available memory such as random access memory (RAM), read only memory (ROM), or any other form of digital storage, local or remote. The memory **708** contains instructions, that when executed by the processor **704** (e.g., central processing unit (CPU), a digital signal processor (DSP), an application-specific integrated circuit (ASIC)), facilitates the operation and processing within of any of the processing chambers illustrated in FIGS. 1-3. The instructions in the memory **708** are in the form of a program product such as a program that implements the method of the present disclosure. The program code of the program product may conform to any one of a number of different programming languages. Illustrative computer-readable storage media include, but are not limited to: (i) non-writable storage media (e.g., read-only memory devices within a computer such as CD-ROM disks readable by a CD-ROM drive, flash memory, ROM chips, or any type of solid-state non-volatile semiconductor memory) on which information is permanently stored; and (ii) writable storage media (e.g., floppy disks within a diskette drive or hard-disk drive or any type of solid-state random-access semiconductor memory) on which alterable information is stored. Such computer-readable storage media, when carrying computer-readable instructions that direct the functions of the methods described herein, are examples of the present disclosure. The program code within memory **708** can be executed by processor **704** in order to perform various processing methods within an apparatus, such as perform one or more of

the method steps described in relation to FIG. 5A or 5B to produce a film stack **400** using the various process chambers and supporting components found in the cluster tool **100** illustrated in FIG. 1.

(76) In one example, the controller **700** may be implemented as the program product stored on a computer-readable storage media (e.g. **708**) for use with a computer system (not shown). The program(s) of the program product define functions of the disclosure, described herein.

(77) Examples disclosed herein are related to an apparatus and method of forming thin film layers on a substrate. While the foregoing is directed to specific examples of the disclosure, other examples may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A method of processing a substrate, comprising: depositing a first piezoelectric material layer on the substrate in a first processing chamber by use of a physical vapor deposition process, wherein the first piezoelectric material layer is formed on the substrate while the substrate is at a first temperature, and a target used in the physical vapor deposition process comprises aluminum and scandium, wherein depositing the first piezoelectric material layer comprises: establishing a distance between a surface of the target and a surface of the substrate, the distance being between about 30 mm and 70 mm; maintaining a bias power between about 20 Watts (W) and about 50 W; and maintaining a magnetic rotation speed between about 100 rpm and 150 rpm; after depositing the first piezoelectric material layer, cooling the first piezoelectric material layer and the substrate in a second processing chamber; and after cooling the first piezoelectric material layer and the substrate, depositing a second piezoelectric material layer on the first piezoelectric material layer at a second temperature, wherein: the first piezoelectric material layer and the second piezoelectric material layer both comprise a first piezoelectric material, and the first temperature is between 120° C. and 600° C., and the second temperature is less than or equal to 100° C.
2. The method of processing the substrate as recited in claim 1, further comprising: depositing a third piezoelectric material layer on top of the second piezoelectric material layer, wherein the third piezoelectric material layer comprises the first piezoelectric material, and the process of depositing the third piezoelectric material layer comprises depositing the third piezoelectric material layer at a different substrate bias than used while depositing the second piezoelectric material layer, wherein each of the first piezoelectric material layer, the second piezoelectric material layer, and the third piezoelectric material layer is formed by a physical vapor deposition process.
3. The method of processing the substrate as recited in claim 1, further comprising: depositing a third piezoelectric material layer on top of the second piezoelectric material layer, wherein the third piezoelectric material layer comprises the first piezoelectric material, and the process of depositing the third piezoelectric material layer comprises depositing the third piezoelectric material layer at a different pressure used to deposit the second piezoelectric material layer, wherein the second piezoelectric material layer, and the third piezoelectric material layer is performed by use of a physical vapor deposition process.
4. The method of processing the substrate as recited in claim 1, wherein the first piezoelectric material comprises aluminum, scandium, and nitrogen.
5. A method of processing a substrate in a cluster tool, comprising: depositing a seed layer on a substrate in a first process chamber of the cluster tool by use of a physical vapor deposition process, wherein the seed layer is formed on the substrate while the substrate is at a first temperature, and a target used in the physical vapor deposition process comprises aluminum and scandium, wherein depositing the seed layer comprises: establishing a distance between a surface of the target and a surface of the substrate, the distance being between about 30 mm and 70 mm; maintaining a bias power between about 20 Watts (W) and about 50 W; and maintaining a magnetic

rotation speed between about 100 rpm and 150 rpm; after depositing the seed layer, cooling the seed layer and the substrate the seed layer and the substrate in a second processing chamber of the cluster tool; and after cooling the seed layer and the substrate, depositing a first piezoelectric material layer on the seed layer, after cooling the substrate to a second temperature, wherein: the seed layer and the first piezoelectric material layer both comprise a first material, and the first temperature is between 120° C. and 600° C., and the second temperature is less than or equal to 100° C.

6. The method of processing the substrate as recited in claim 5, further comprising: exposing a surface of the substrate to a plasma before depositing the seed layer.

7. The method of processing the substrate as recited in claim 5, the method further comprises: depositing a second piezoelectric material layer on top of the first piezoelectric material layer, wherein the second piezoelectric material layer comprises the first material, and the process of depositing the second piezoelectric material layer comprises depositing the second piezoelectric material layer at a different substrate bias than used while depositing the first piezoelectric material layer, wherein each of the seed layer, the first piezoelectric material layer, and the second piezoelectric material layer is formed by a physical vapor deposition process.

8. The method of processing the substrate as recited in claim 5, the method further comprises: depositing a second piezoelectric material layer on top of the first piezoelectric material layer, wherein the second piezoelectric material layer comprises the first material, and the process of depositing the second piezoelectric material layer comprises depositing the second piezoelectric material layer at a different pressure or substrate bias than used to deposit the first piezoelectric material layer, wherein the process of depositing the seed layer and the first piezoelectric material layer is performed by use of a physical vapor deposition process performed.
